

Assignment 3: ECO764

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Question 1: Option Strategies for May 15, 2009

1. Political and Market Scenario

On May 15, 2009, the Indian capital market was characterized by extreme uncertainty regarding the world's largest electoral process. Raj Sharma, a senior research analyst, noted that market trajectory remained unpredictable due to the upcoming election results scheduled for announcement on May 16.

- **Market Sentiment:** Psychological forces of greed, fear, and hope inhabited the market as participants awaited "D-Day".
- **Objective:** To craft optimal strategies by the Close of Business (COB) on May 15 to capitalize on the expected "gap up" or "gap down" opening on Monday.
- **Underlying Price:** The CNX Nifty Index underlying value was 3,671.65.

2. Formulated Option Strategies

Based on the Nifty European Options data from **Exhibit 3**, the following strategies are designed to handle the anticipated high volatility.

Strategy A: Long Straddle (Aggressive/Volatile)

This strategy involves buying an At-The-Money (ATM) call and put to profit from a massive swing in either direction.

- **Execution:** Buy 3700 Call (150.00) and 3700 Put (162.00).
- **Net Outlay (Max Risk):** 312.00 points.

Strategy B: Bull Call Spread (Defensive/Bullish)

A defensive strategy providing higher profit potential while limiting the risk of total premium loss.

- **Execution:** Buy 3700 Call (150.00) and Sell 3800 Call (108.00).
- **Net Outlay (Max Risk):** 42.00 points.

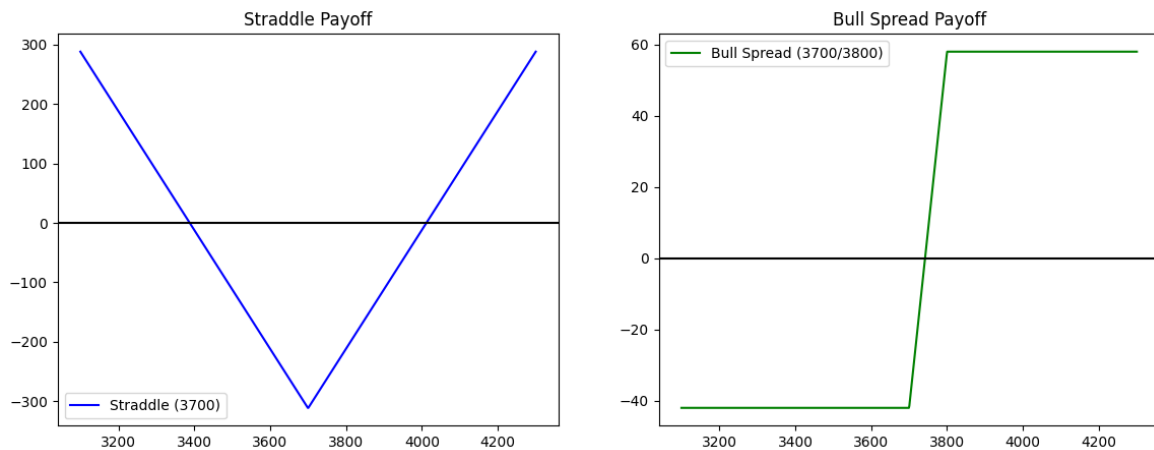


Figure 1: Strategic Payoff Diagrams established on May 15, 2009

Question 2: Evaluation of Strategic Performance Potential

Using the data provided in **Case Exhibit 3**, the performance of the chosen strategies is evaluated through liquidity analysis, breakeven thresholds, and risk-reward profiles.

1. Liquidity and Market Consensus Analysis

An evaluation of the Open Interest (OI) in Exhibit 3 reveals significant market activity at the chosen strikes, ensuring hedging effectiveness and ease of exit.

- **Strike 3700:** The 3700 Call and Put strikes show a combined Open Interest of over 6.3 million contracts (3,258,350 for Calls and 3,136,550 for Puts), confirming it as the primary battleground for market expectations.
- **Resistance Levels:** A peak in Call OI at the 4000 strike (4,078,500) suggests that while traders are bullish, there is a strong expectation of resistance at the 4000 level.

2. Risk-Reward and Breakeven Evaluation

The performance potential of each strategy depends on its respective breakeven points relative to the spot price of 3,671.65.

Strategy	Max Risk	Lower Breakeven	Upper Breakeven	Strategic Class
Long Straddle	312.00 pts	3,388.00	4,012.00	Aggressive [cite: 113]
Bull Call Spread	42.00 pts	3,742.00	N/A	Defensive [cite: 110]

Table 1: Ex-ante Performance Metrics (Nifty Index)

3. Comparative Conclusion

The **Long Straddle** requires a minimum price movement of $\pm 8.5\%$ to reach profitability, making it highly dependent on a massive electoral swing. In contrast, the **Bull Call Spread** reduces the capital commitment by 86.5% compared to the straddle, requiring only a 1.9% upward move to reach its breakeven, aligning with Sharma’s defensive recommendation for retail clients seeking ”exponential profits” without ”losing significant money”.

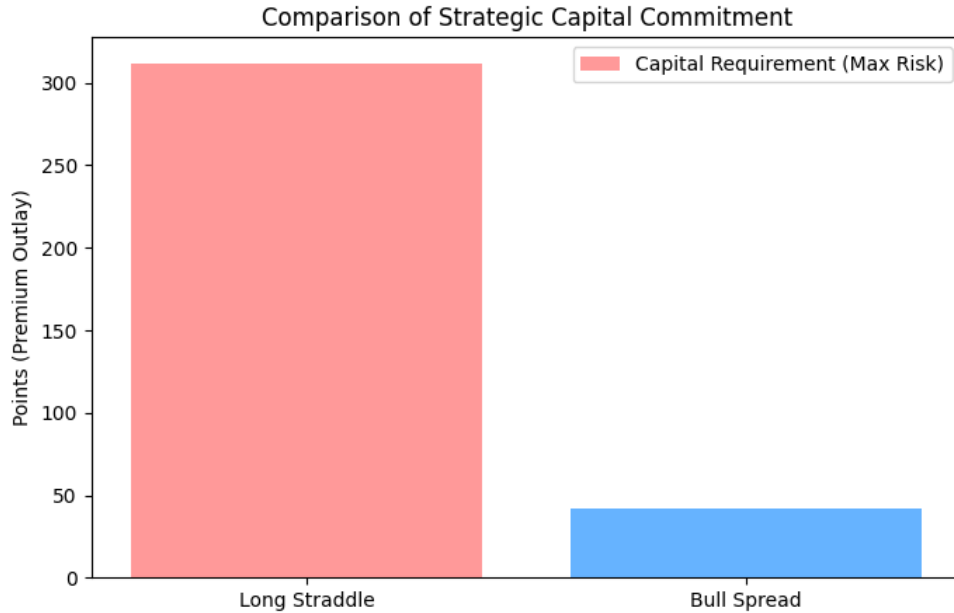


Figure 2: Performance Evaluation: Comparative Capital Commitment and Risk Levels

Question 3: Hedging Effectiveness of Stock Options

1. Hedging Strategy: The Protective Put

To examine the hedging effectiveness, a **Protective Put** strategy is employed for both ICICI Bank and Reliance. This involves holding the underlying stock and purchasing an At-The-Money (ATM) Put option. This strategy provides a floor for the portfolio, protecting against the high probability of a market crash (lower circuit) during the 2009 election results.

2. Quantitative Analysis for ICICI Bank and Reliance

Based on the data provided in the option chains for May 15 and May 19, the hedging performance is calculated as follows:

- **ICICI Bank (Strike 560):**

- Cost of Hedged Position (May 15): $574.70 \text{ (Spot)} + 28.70 \text{ (Put)} = 603.40$.
- Value on May 19: $756.15 \text{ (Spot)} + 2.50 \text{ (Put)} = 758.65$.
- **Hedged Profit:** 155.25 per share (vs. 181.45 unhedged).

- **Reliance (Strike 1950):**

- Cost of Hedged Position (May 15): $1950.70 \text{ (Spot)} + 85.35 \text{ (Put)} = 2036.05$.
- Value on May 19: $2230.90 \text{ (Spot)} + 15.75 \text{ (Put)} = 2246.65$.
- **Hedged Profit:** 210.60 per share (vs. 280.20 unhedged).

3. Comparative Performance and Effectiveness

Stock	Unhedged Return	Hedged Return	Cost of Hedge	Downside Protection
ICICI Bank	31.57%	25.73%	5.84%	Strike 560 (Lower Floor)
Reliance	14.36%	10.34%	4.02%	Strike 1950 (Lower Floor)

Table 2: Comparison of Hedging Effectiveness (May 15 vs May 19, 2009)

4. Conclusion on Hedging Effectiveness

The hedging effectiveness of the American options in this scenario is demonstrated by the trade-off between return and risk:

1. **Downside Truncation:** While the market surged, the hedge ensured that if a negative "hung parliament" outcome had occurred, losses would have been capped at the strike prices.
2. **Cost of Insurance:** The "Cost of Hedge" (the difference between unhedged and hedged returns) represents the insurance premium paid for market uncertainty. For Reliance, this cost was 4.02%, while for ICICI Bank, it was 5.84%, reflecting the higher implied volatility of the banking sector at that time.
3. **Effectiveness:** The options were highly effective as risk management tools, as they allowed participants to stay invested in the equity market during an extreme macro-financial event while maintaining a predefined maximum loss limit.

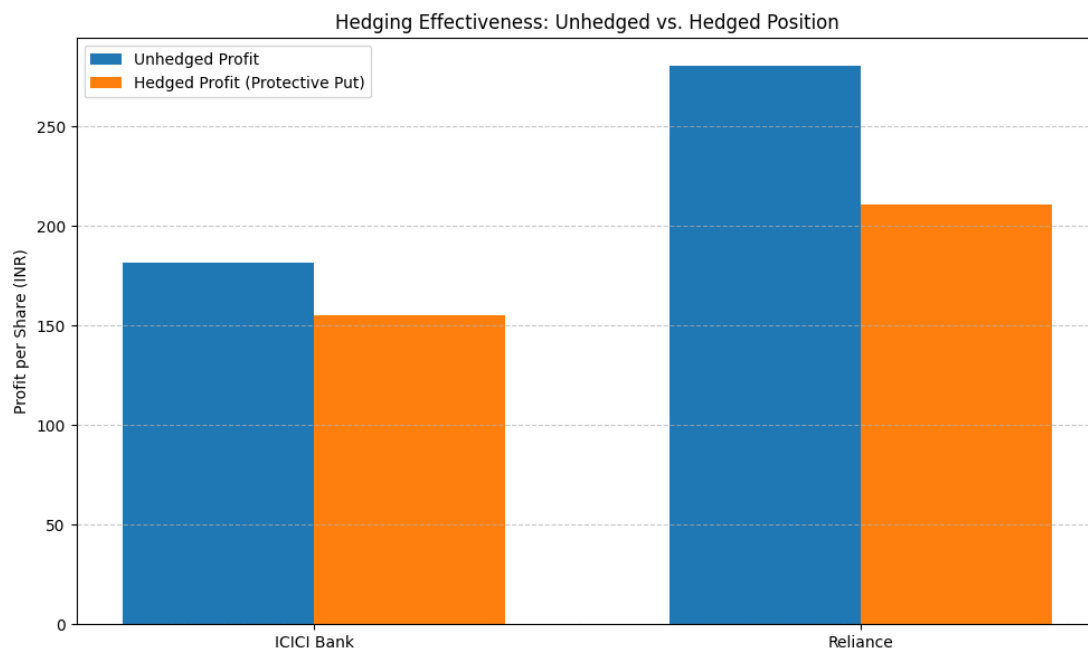


Figure 3: Comparative Analysis: Profit Potential of Hedged vs. Unhedged Positions

Question 4: Analysis of Return Potential Across Market Sectors

1. Comprehensive Evaluation of Exhibit 5

The examination of Exhibit 5, encompassing 47 key stocks, reveals that the return potential was not uniform across the capital market. While the broader Nifty index witnessed an appreciation of 17.33%, individual stock performance displayed a significant standard deviation, ranging from -1.6% to 46.1% .

- **Top Performers (Beta Sensitivity):** High-beta cyclical sectors like Construction (DLF: 46.1%), Engineering (LT: 36.4%), and Banking (Kotak Bank: 32.9% , SBI: 32.4%) surged as the political risk premium was removed.
- **Defensive Laggards:** Stocks in Pharmaceuticals (Cipla: -1.6%) and FMCG (HUL: 2.4% , ITC: 2.6%) significantly underperformed the index.

2. Reasons for Divergence

The divergence in performance between May 15 and May 19, 2009, highlights that stock returns are influenced by sectoral specificities:

1. **Sectoral Beta:** Cyclical industries have higher sensitivity to macro-financial events like elections. A stable government forecast led to immediate valuation re-ratings for these

sectors.

2. **Asset Rotation:** In scenarios of extreme market euphoria, investors rotate capital away from “safe haven” or defensive stocks (Pharma/FMCG) into growth-oriented stocks to capitalize on the rally.
3. **Policy Continuity:** Financial and Infrastructure institutions benefited from the expectation of continued reform momentum, while global-facing sectors (IT) remained muted due to their secondary dependence on domestic politics.

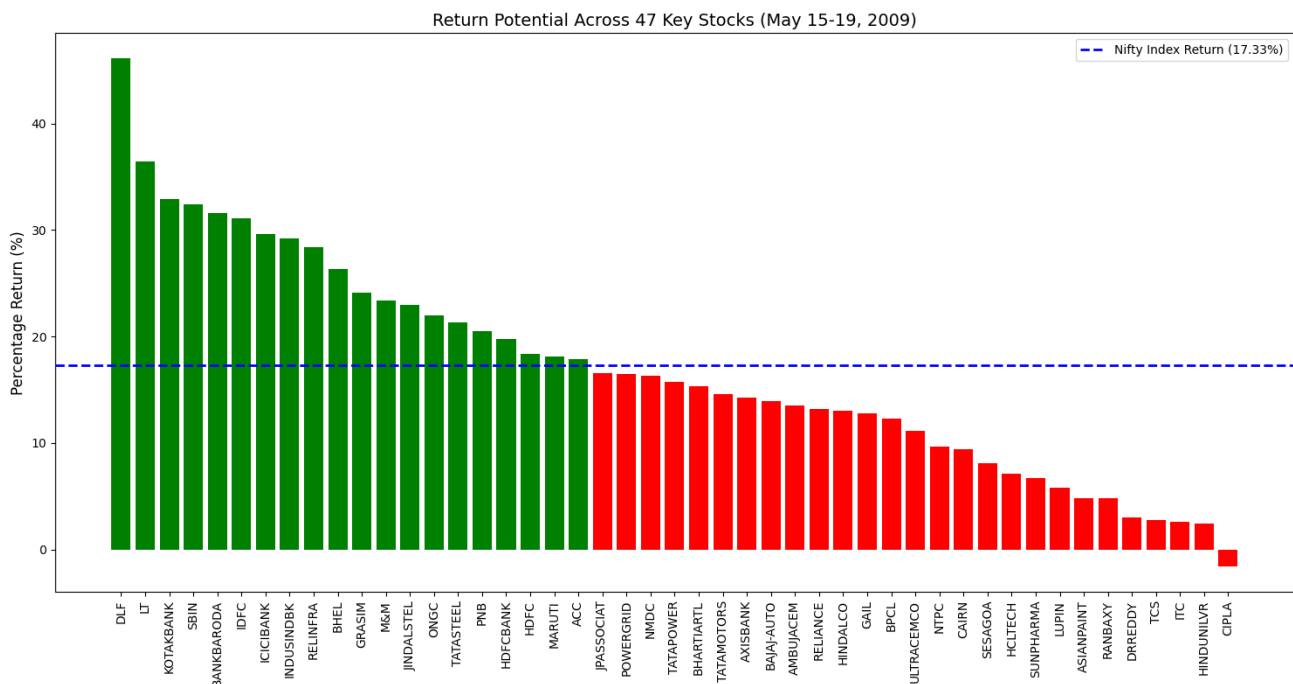


Figure 4: Distribution of Percentage Returns Across 47 Key Stocks (May 15-19, 2009)

Question 5: Option Strategies for Lower Circuit and Extreme Volatility

1. Historical Benchmark: Analysis of Exhibit 2 (Jan 16–29, 2008)

The historical data in Exhibit 2 provides a template for a market crash scenario. Over 10 trading sessions, the Nifty Index experienced a maximum peak-to-trough decline of 26.6% (from 6,065.00 to 4,448.50). This environment was characterized by:

- **Accelerated Momentum:** The most severe selling occurred between Jan 18 and Jan 21, where the index lost nearly 500 points in the closing figures.
- **Extreme Intraday Range:** On Jan 22, the difference between the high and low was 754.85 points, indicating that “Lower Circuit” conditions were met.

- **Liquidity Surge:** Turnover spiked to over 125,000 million during the crash, reflecting panic-driven exit and speculative entry.

2. Formulated Option Strategies (Based on Exhibit 3)

In light of the Exhibit 2 scenario, the following strategies are formulated using premiums from May 15 (**Exhibit 3**) to protect against a similar plunge.

Strategy A: Aggressive Bearish (Long Put)

Designed to capitalize on a “Lower Circuit” plunge with unlimited profit potential.

- **Execution:** Buy one 3600 Strike Put at 119.65 points.
- **Breakeven:** Nifty Index at 3,480.35.

Strategy B: Defensive Bearish (Bear Put Spread)

Designed for high-volatility environments to reduce the net cost of insurance.

- **Execution:** Buy 3600 Put (119.65) and Sell 3400 Put (63.00).
- **Net Cost:** 56.65 points.

Strategy C: Volatility Play (Long Straddle)

Recommended for the extreme direction-neutral volatility seen in the recovery phase of Jan 22–23.

- **Execution:** Buy 3700 Call (150.00) and Buy 3700 Put (162.00).
- **Net Cost:** 312.00 points.

Scenario	Strategy	Net Cost (Pts)	Risk Level	Rationale
Sudden Plunge	Long Put (3600)	119.65	High	Unlimited Downside Profit
Market Halt/Circuit	Bear Put Spread	56.65	Moderate	Cost-Efficient Hedge
Extreme Volatility	Long Straddle (3700)	312.00	High	Direction-Neutral Bet

Table 3: Formulated Strategies for Scenarios Equivalent to Exhibit 2

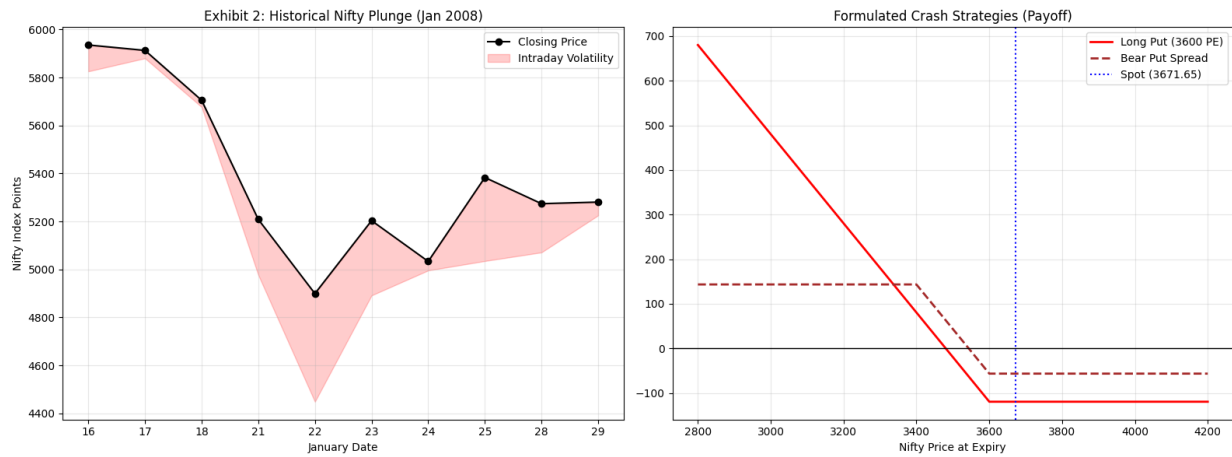


Figure 5: Plot: Historical Volatility Analysis (Jan 2008) and Crash Strategy Payoffs

References

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- [5] Hull, J. C. (2022). *Options, Futures, and Other Derivatives*. Pearson Education. (Standard resource for Option Strategy Payoffs and Hedging Theory).